

# Arunabha Majumder

## Mechanical Engineer — Design Optimisation, Continuous Improvement & Testing

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### SUMMARY

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Mechanical engineer who improves products through testing, design optimisation and clear documentation. Improved a mechatronic system's performance by more than 25% through design changes, and enjoys solving practical problems with production-minded, cross-functional teams. Strong in CAD (SolidWorks, CWSA) and eager to work in PLM systems.

### KEY RESULTS

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- **>25%** performance gain through design optimisation
- **<10%** RMS error on a verified measurement method
- **Concept → test → improve:** a repeated continuous-improvement loop on real hardware

### EXPERIENCE

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#### Research Engineer – Mechanical Design & Continuous Improvement (PhD Project: VIEXO) | 2023 – Present

*Aalborg University, Aalborg, Denmark*

- Improved performance by more than 25% (energy consumption, measured in lab tests) by acting as the mechanical expert on a shoulder exoskeleton and making design modifications and optimisations to its actuation system, a parallel spring working with a 6 Nm motor (Best Paper Award, IFToMM ISRM 2026).
- Increased reliability through repeated test campaigns by running a continuous improvement loop: testing prototypes, finding the cause of mechanical, electronic and control issues, and implementing design changes in SolidWorks.
- Kept designs buildable and well documented by maintaining 3D models, assemblies, technical drawings and product documentation for machined and 3D-printed parts, using FEA for strength, stiffness and weight.
- Verified a force-estimation method to below 10% RMS error against a force sensor by developing it for a 5-bar planar parallel robot using variable-stiffness actuators and the robot Jacobian.
- Delivered on multiple parallel tasks (design, testing, a 12-participant data collection, and supervising Bachelor's students in 2023 – 2025) by working in a structured way and collaborating across functions (mechanical, electronics, control, biomechanics).

#### MTech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022

*CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Reduced position-tracking error to 0.3–0.78% by iteratively testing and improving the control of a non-linear pneumatic actuator, outperforming classical PID.
- Built a reliable test set-up by fabricating pneumatic artificial muscles in-house and integrating valves, pressure sensors and DAQ.

### SKILLS

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- **Mechanical design & improvement:** Design modifications, optimisation, FEA, DFA, technical drawings, documentation
- **CAD & PLM:** SolidWorks (CWSA), Fusion 360; eager to learn PLM and engineering change processes
- **Testing & troubleshooting:** Test rigs, lab testing, finding the cause of issues, verification
- **Mechatronics:** Motors, sensors, embedded control, MATLAB/Simulink, Python
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

## EDUCATION

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**Ph.D. Candidate, Mechatronics & Robotics** | 2023 – Present

*Aalborg University, Department of Materials and Production, Aalborg, Denmark*

**M.Tech, Mechatronics** | 2020 – 2022

*AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

**B.Tech, Mechanical Engineering** | 2015 – 2019

*Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India*

## HANDS-ON PROJECTS

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- Rehabilitation exoskeleton (Best Major Project Award and TATA Technologies Innovation Grant, 2019), Go-Kart built from the ground up (2017), Efficycle human-electric hybrid vehicle (2018)

## AWARDS & PUBLICATIONS

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- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Two peer-reviewed publications: ISRM 2026 (Springer) and IEEE CONECCT 2022

## REFERENCES

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Available upon request.